



Why AI?



Economic Impact

1. Job Creation and Transformation:

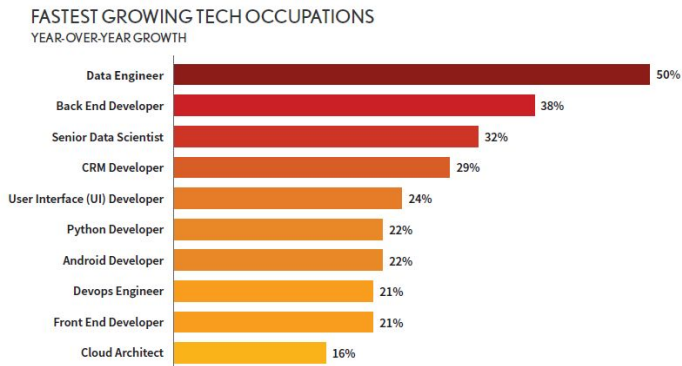
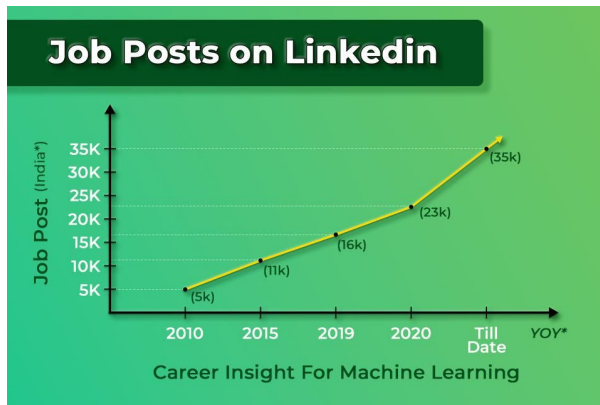
- AI is transforming industries by automating routine tasks, increasing efficiency, and creating new job opportunities in fields such as data analysis, machine learning engineering, and AI ethics.

2. Productivity and Innovation:

- AI technologies can significantly enhance productivity by optimizing processes, reducing errors, and enabling the development of innovative products and services.

3. Competitive Advantage:

- Organizations that effectively leverage AI can gain a competitive edge by making more informed decisions, improving customer experiences, and predicting market trends.



Social Impact

1. **Improving Quality of Life:**

- AI is being used to improve healthcare through predictive analytics which can lead to better patient outcomes and more efficient healthcare systems.

2. **Addressing Global Challenges:**

- AI can help tackle pressing global issues such as climate change, food security, and disaster response by analyzing large datasets to provide insights and solutions.

3. **Enhanced Accessibility:**

- AI-driven technologies like speech recognition, language translation, and assistive devices are helping to bridge accessibility gaps, making technology more inclusive for people with disabilities.

Ethical and Regulatory Considerations

1. **Ethical AI Development:**

- Understanding AI is crucial to ensure that it is developed and deployed ethically, addressing issues such as bias, fairness, transparency, and accountability.

2. **Policy and Regulation:**

- Policymakers need a thorough understanding of AI to create regulations that protect public interest while fostering innovation and preventing misuse.

3. **Privacy and Security:**

- Studying AI helps address concerns related to data privacy and security, ensuring that AI systems are designed to protect sensitive information and prevent malicious attacks.

Technological Advancement

1. **Driving Research and Development:**

- AI is at the forefront of technological innovation, pushing the boundaries of what is possible in fields like robotics, natural language processing, and computer vision.

2. **Interdisciplinary Collaboration:**

- AI research fosters collaboration across various disciplines, including computer science, neuroscience, linguistics, and psychology, leading to a more comprehensive understanding of intelligence and cognition.

3. **Preparing for the Future:**

- As AI continues to evolve, it is essential to prepare for its future implications, including the potential development of artificial general intelligence (AGI) and its societal impact.

Societal and Ethical Considerations

1. **Human-AI Interaction:**

- Understanding how humans and AI can interact effectively is vital for developing systems that augment human capabilities and provide intuitive user experiences.

2. **Long-term Implications:**

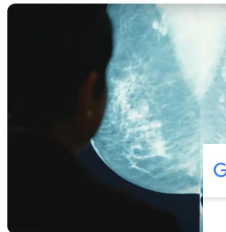
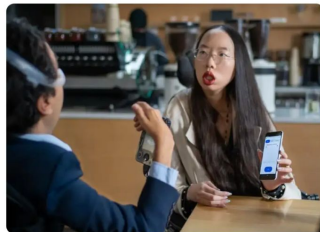
- Studying AI helps society anticipate and plan for long-term implications, such as changes in the labor market, shifts in power dynamics, and the potential for superintelligent AI.

3. **Educational Transformation:**

- AI is revolutionizing education through personalized learning experiences, intelligent tutoring systems, and data-driven insights into student performance.

Our Principles

While we are optimistic about the potential of AI, we recognize that advanced technologies can raise important challenges that must be addressed clearly, thoughtfully, and affirmatively. These AI Principles describe our commitment to developing technology responsibly and work to establish specific application areas we will not pursue.



Objectives for AI applications

1. Be socially beneficial.



2. Avoid creating or reinforcing unfair bias.



3. Be built and tested for safety.



4. Be accountable to people.



5. Incorporate privacy design principles.



6. Uphold high standards of scientific excellence.



7. Be made available for uses that accord with these principles.



<https://ai.google/responsibility/principles/>



ACL Rolling Review

An initiative of the Association for Computational Linguistics

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Members of the ACL are responsible for adhering to the [ACL code of ethics](#). The ARR Responsible NLP Research checklist is designed to encourage best practices for responsible research, addressing issues of research ethics, societal impact and reproducibility. It is largely based on the [NeurIPS 2021 paper checklist](#), the reproducible data checklist from [Rogers, Baldwin, Leins's EMNLP 2021 paper "Just What do You Think You're Doing, Dave? A Checklist for Responsible Data Use in NLP"](#), and the NLP Reproducibility Checklist built from [Dodge et al. EMNLP 2019's paper "Show Your Work: Improved Reporting of Experimental Results"](#).

We expect authors to show that they follow best practices in two ways:

1. by filling in the checklist to ensure that best practices are put in place when using, creating or providing assets,
2. by including a discussion in the paper about any potential positive or negative societal impacts and any limitations of the work. The guidelines below provide additional information about what should be discussed.

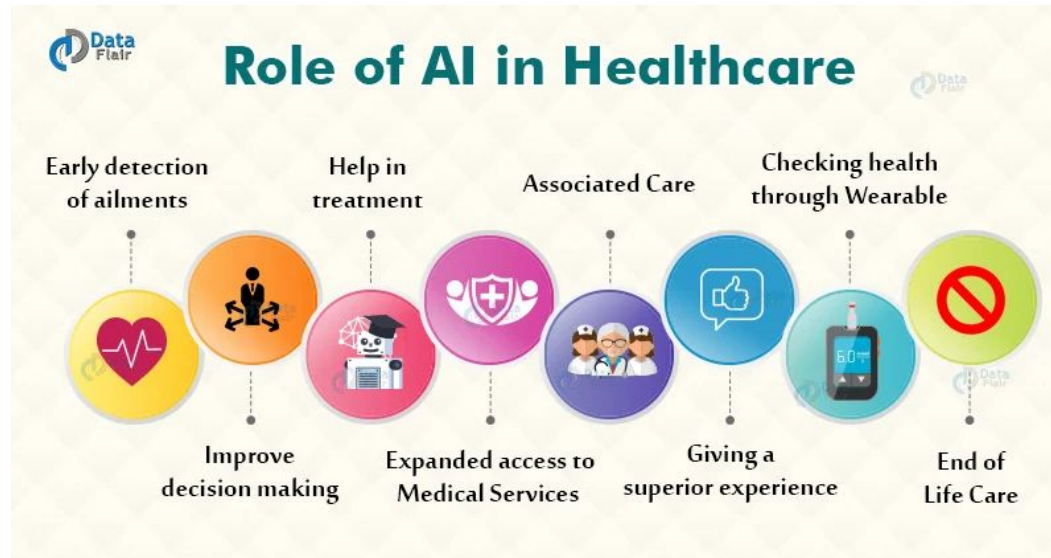
<https://aclanthology.org/2021.findings-emnlp.414.pdf>

<https://aclanthology.org/D19-1224.pdf>

Applications

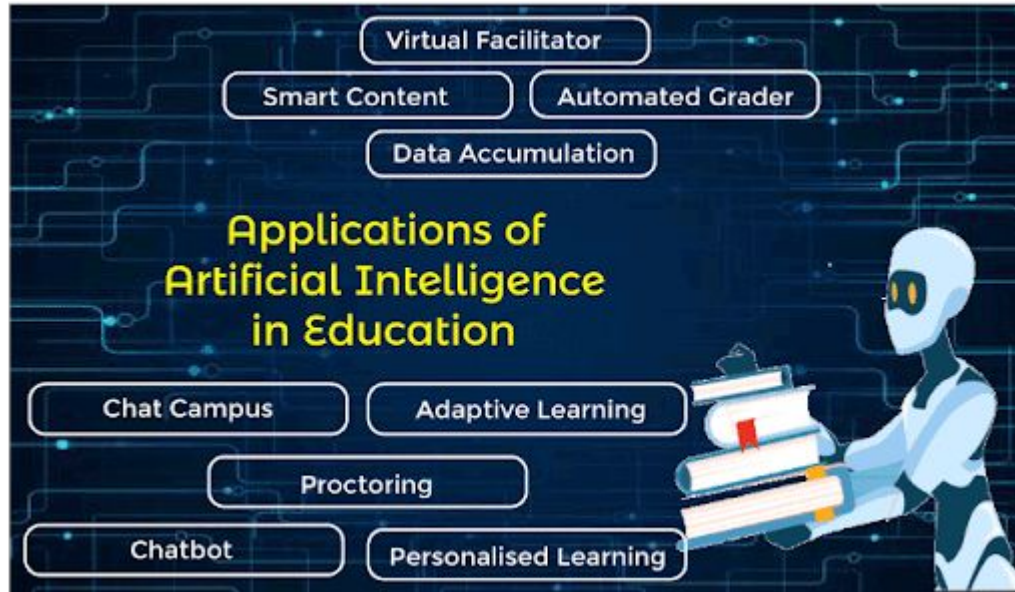
1. Healthcare

- **Medical Diagnosis:** AI systems can analyze medical images and data to help diagnose diseases like cancer, heart disease, and neurological disorders.
- **Personalized Medicine:** AI can tailor treatments to individual patients based on their genetic makeup and health history.



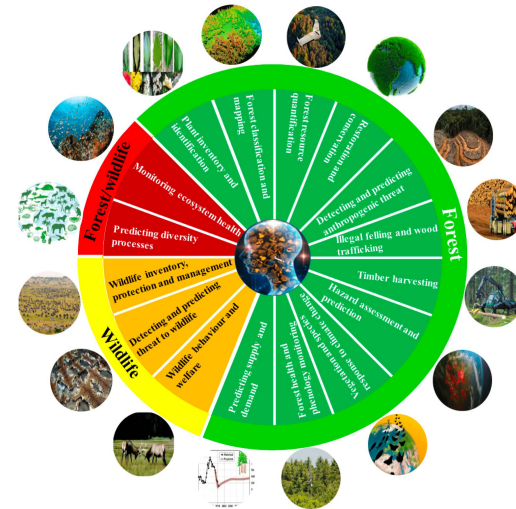
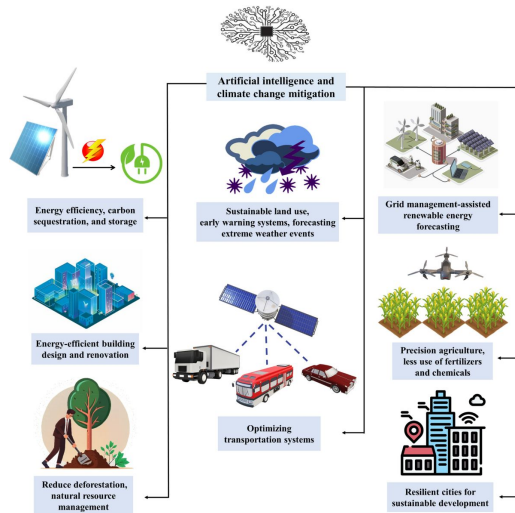
2. Education

- **Personalized Learning:** AI-driven platforms can customize learning experiences based on students' strengths, weaknesses, and learning styles.
- **Automated Grading:** AI can assist teachers by grading assignments and providing feedback, allowing them to focus more on instruction.



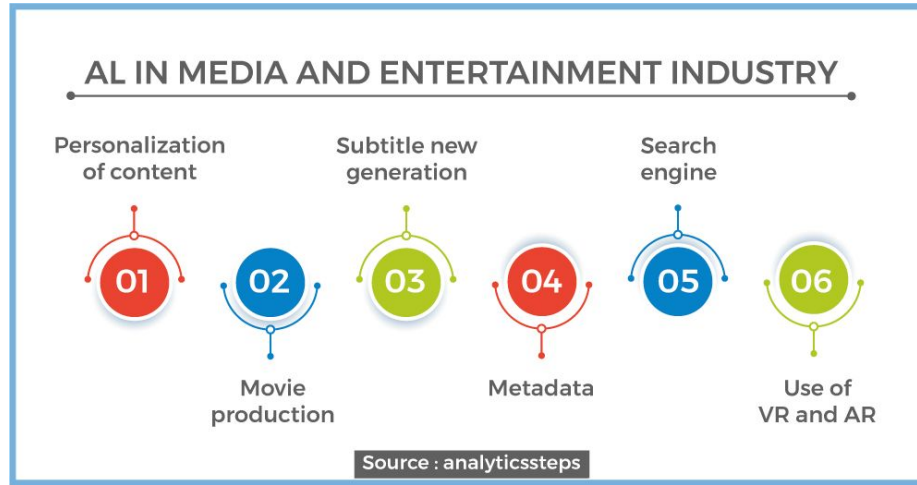
3. Environment and Sustainability

- **Climate Change Modeling:** AI can analyze large datasets to predict climate changes and model the impacts of various environmental policies.
- **Wildlife Conservation:** AI is used to monitor wildlife populations and combat poaching through image recognition and data analysis.

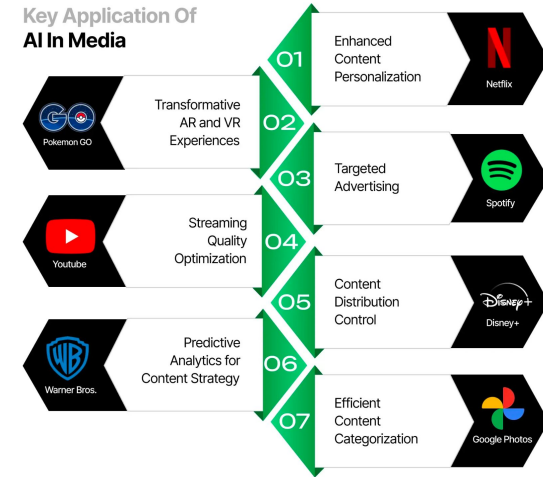


4. Entertainment and Media

- **Content Creation:** AI can generate music, art, and stories, showcasing creative potentials that appeal to students interested in the arts.
- **Game Development:** AI is integral in creating intelligent behaviors in non-player characters (NPCs) and enhancing gaming experiences.

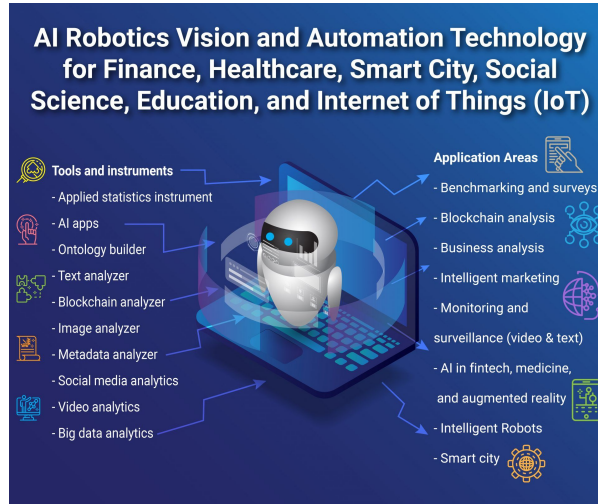


Key Application Of AI In Media



5. Robotics

- **Autonomous Vehicles:** AI powers self-driving cars, which can revolutionize transportation and logistics.
- **Drones:** AI-driven drones are used for delivery, agriculture, and disaster relief efforts.



APPLICATION OF AI IN ROBOTICS

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1

INDUSTRY

Robots are used in industries as an important individual organ which plays a vital role in productivity to perform an assigned task and whole process is being managed by these artificial intelligence machines.

2

MILITARY

An Artificial Intelligent Robot namely "Daksh" developed by "Defense Research and Development Organization (DRDO)" and its functionality is to remove the life threatening objects in danger zone.

3

MEDICINE

Many complicated operations like brain tumor etc. are being performed by the modern artificial intelligence Robots which shows the advancement in medical fields.

4

ENTERTAINMENT

The science fiction movies are full of artificial intelligence robots to illuminate the advancement in entertainment.

5

AEROSPACE

The discovery of new planets, the working of satellites, and sending modern aeronautical jets towards space to discover new things in space is not possible without artificial intelligence robotic technology.

6

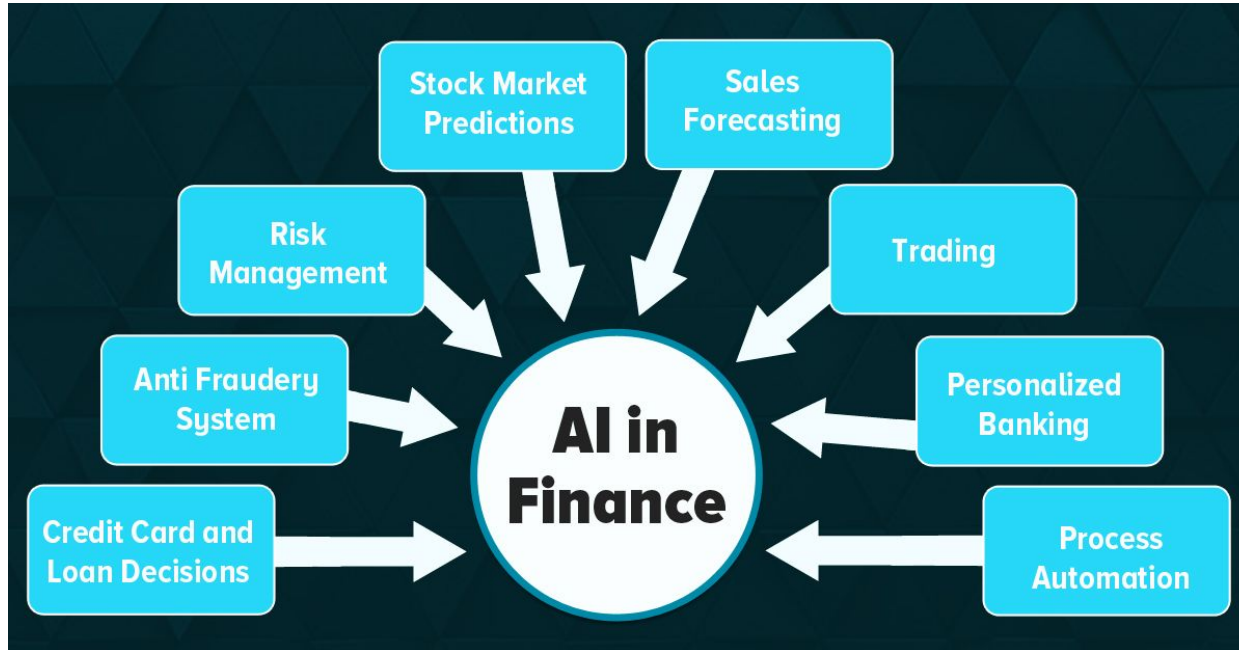
SCIENTIFIC LABS

The danger to human life is reduced where the AI robots assist humans in the laboratories to perform sensitive scientific experiments.

DatabaseTown.com

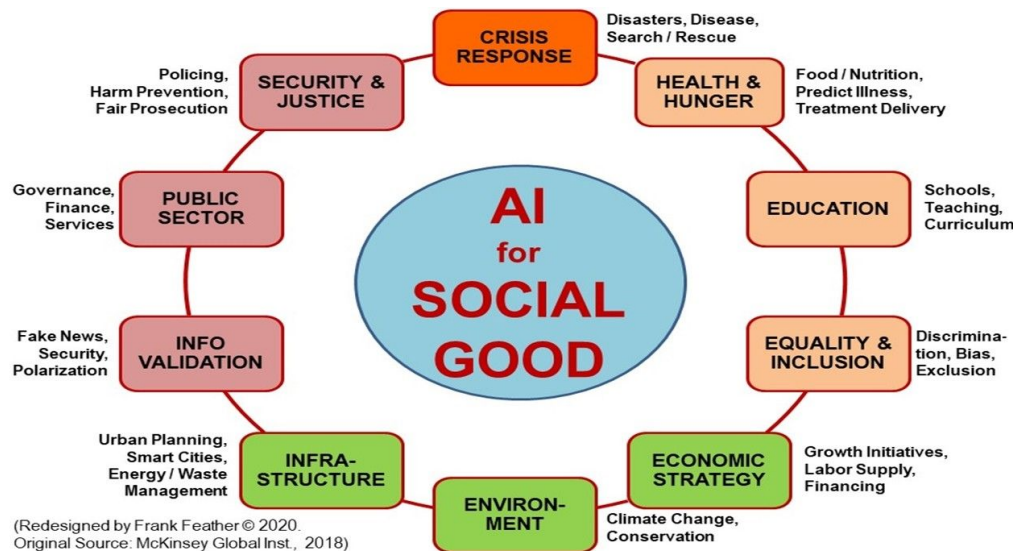
6. Finance

- **Fraud Detection:** AI systems can detect fraudulent transactions by analyzing patterns in financial data.
- **Algorithmic Trading:** AI algorithms can analyze market trends and make trading decisions in real-time.



7. Social Good

- **Disaster Response:** AI helps in predicting natural disasters and coordinating relief efforts by analyzing data from various sources.
- **Humanitarian Aid:** AI systems can optimize resource distribution in crisis situations, ensuring that aid reaches those in need efficiently.



8. Daily Life Enhancements

- **Smart Assistants:** AI-powered virtual assistants like Siri, Alexa, and Google Assistant can help with everyday tasks and information retrieval.
- **Home Automation:** AI systems control smart homes, managing lighting, security, and climate to improve comfort and energy efficiency.

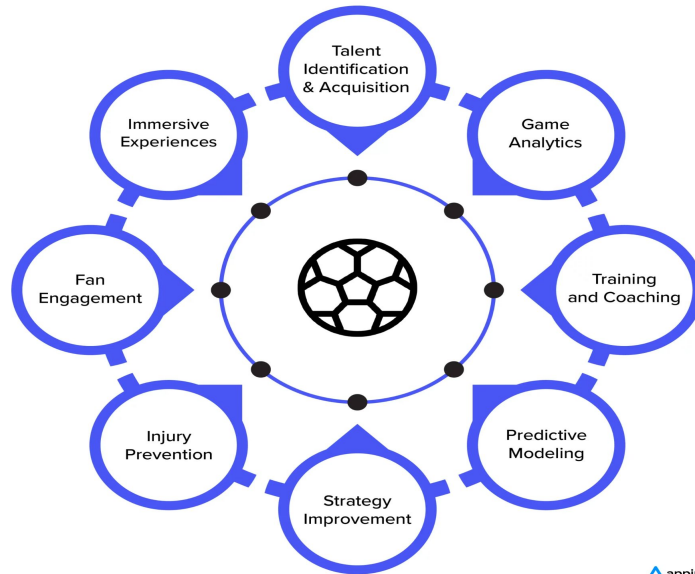


Horus, Aira and BeMyEyes for visually-impaired patients	KidsMD through Amazon Alexa for parents of small children	Other Apps, Other Applications
• Reads text • Recognizes objects and faces • Identifies obstacles on patients' path	• Educates user on symptoms of common illnesses • Recommends remedial solutions • Schedules doctor appointments	• Monitors food intake for diabetics, makes personalized recommendations on what/how much to eat or drink, and provides reminders to measure blood glucose • Assists patients in booking rides to and from the hospital via ride-sharing services • Explains discharge instructions to patients for post-surgery care

9. Sports

- **Performance Analysis:** AI can analyze athletes' performance and suggest improvements based on data-driven insights.
- **Fan Engagement:** AI enhances the fan experience through personalized content and interactive features during live events.

Artificial Intelligence Framework for Sports



10. Scientific Research

- **Drug Discovery:** AI accelerates the process of discovering new drugs by analyzing vast amounts of biological data.
- **Astronomy:** AI helps in processing data from telescopes and discovering new celestial bodies and phenomena.

